



Guideline

Thiamine Treatment for Alcohol Dependent Patients

Scope

Site	Service/Department/Unit	Disciplines
SCGOPHCG	Hospital-Wide	Medical, Nursing, Pharmacy

Introduction

Thiamine is a water-soluble vitamin (B1) essential for several enzymatic processes associated with energy metabolism. The recommended daily intake of thiamine is 1.1-1.4mg/day in adults. The total body stores of thiamine are approximately 30-50mg which can be depleted within 4-6 weeks if oral intake is insufficient.

Chronic consumption of alcohol can result in thiamine deficiency due to insufficient dietary intake, decreased absorption, and impaired utilization, ultimately leading to various consequences, notably Wernicke's encephalopathy (WE). Timely intervention in treating Wernicke's encephalopathy is vital due to its potentially life-threatening nature and the potential for irreversible neurological harm. Prompt administration of thiamine supplementation can aid in reversing or halting the progression of neurological decline, minimising the likelihood of life-long complications such as Korsakoff syndrome, which profoundly impairs memory and cognitive functions.

There is limited clinical research and consensus on the optimal dosage and duration of thiamine supplementation in alcohol dependent patients. Consequently, treatment often involves administering higher doses of thiamine intravenously (as oral bioavailability is low) to ensure sufficient levels are reached, followed by continued oral supplementation. Individual patient factors and clinical judgment are crucial in tailoring the appropriate dosage regimen for each case.

Signs & Symptoms

Wernicke's Encephalopathy	Suspect when person has one of more of: altered mental state, decreased consciousness, ataxia, memory disturbance, coma, ophthalmoplegia, nystagmus, unexplained hypotension or hypothermia
Acute Alcohol Withdrawal	Anxiety, tremor, nausea and vomiting, sweating, hallucinations, agitation, fever without infection, tachycardia, hypertension, tachypnoea, insomnia, and seizures in severe cases.
Korsakoff's Psychosis	Severe anterograde and retrograde amnesia, confabulation, disorientation, attention deficits, and executive function impairments.

Contraindications

- Known hypersensitivity or anaphylaxis to any of the ingredients.



Thiamine Administration in Alcohol-Related Conditions

Dosage

Clinical Considerations



Identify Hypersensitivity & Anaphylactic reactions

This can rarely occur - Manage with standard protocol



Check Serum Magnesium

Measure magnesium levels and replace if low



Administer Thiamine before Glucose

Give thiamine before any supplemental glucose - Glucose can further deplete thiamine stores & precipitate Wernicke's Encephalopathy



Consider Further Management & Drug & Alcohol Team Involvement

Switching from IV to PO regimen

No Evidence of Chronic Alcohol Misuse

Thiamine given at clinician's discretion

Alcohol Intoxication with suspected chronic Alcohol abuse

Thiamine 300 mg IV daily

Acute Alcohol Withdrawal without altered mental state

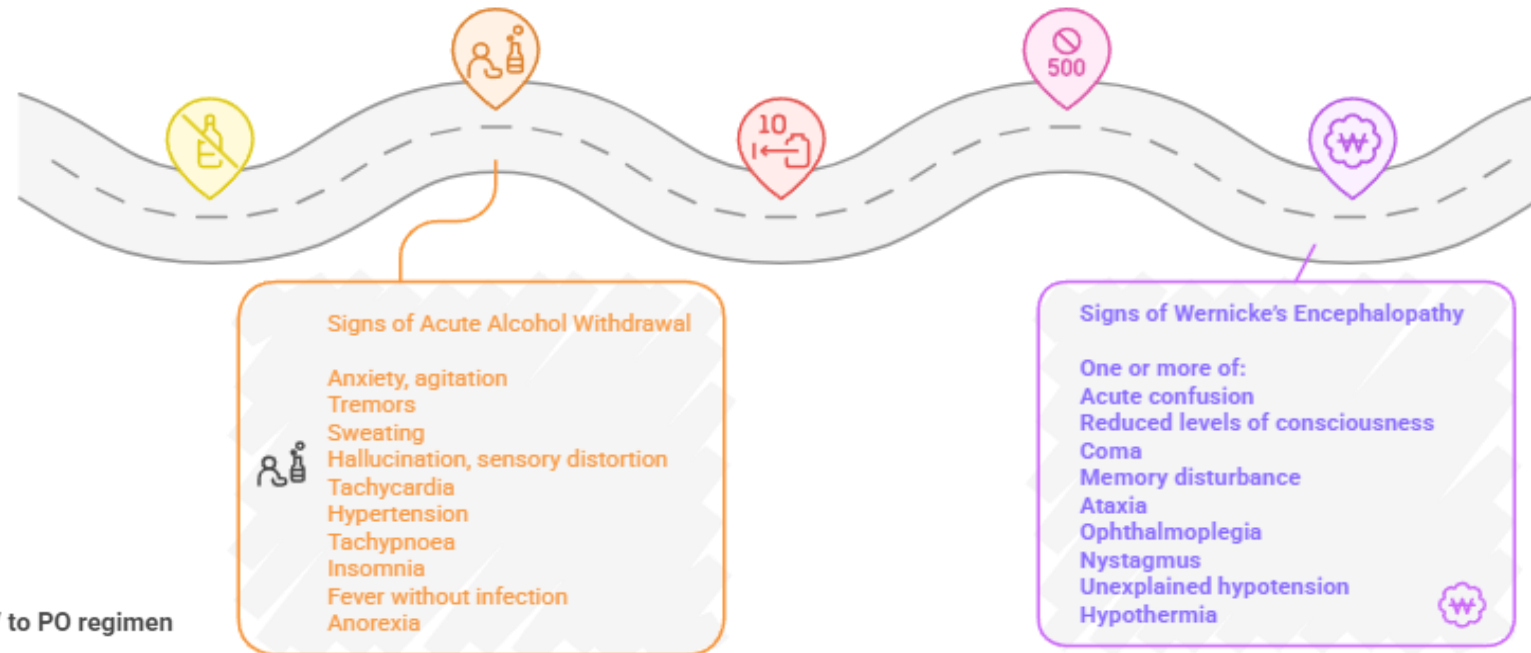
Thiamine 300 mg IV TDS for 24 hours then review

Altered Mental State with suspected chronic alcohol abuse

Thiamine 300mg IV TDS for 3 days or until mental state normalises

High Suspicion of Wernicke's Encephalopathy

Consider Thiamine 500 mg IV TDS for 3 days or until mental state normalises



Signs of Acute Alcohol Withdrawal



Anxiety, agitation
Tremors
Sweating
Hallucination, sensory distortion
Tachycardia
Hypertension
Tachypnoea
Insomnia
Fever without infection
Anorexia

Signs of Wernicke's Encephalopathy

One or more of:
Acute confusion
Reduced levels of consciousness
Coma
Memory disturbance
Ataxia
Ophthalmoplegia
Nystagmus
Unexplained hypotension
Hypothermia



Initiate PO Thiamine Regimen



Administer 100mg PO TDS



Monitor Sobriety



Continue for One Month



Switch to 100mg PO Daily ongoing



Acknowledgments

We acknowledge the following site endorsed guidelines used in preparation of this guideline.

- Fiona Stanley Hospital – Thiamine deficiency: Treatment for alcohol dependent patients (Clinical Guideline FSH-ED-GUI-0004), March 2020.
- Fiona Stanley Hospital – Thiamine (FSH Pharmacy Department – Intravenous Drug Administration Guideline), April 2024.
- Royal Perth Hospital - Guideline for Thiamine Administration in the Emergency Department (Flowchart), October 2021.

Related Documents

- For information regarding thiamine supplementation for oral and enteral refeeding syndrome please refer to the Oral and Enteral Refeeding Syndrome Guideline, November 2019
- For information regarding thiamine supplementation for refeeding in eating disorder patients, please refer to the WA Eating Disorders Outreach & Consultation Service (WAEDOCs) Eating Disorders: the Management of Youth and Adults - a Quick Reference Guide, July 2020

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